Cap and Escape in Trade Agreements*

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November 10, 2014

Abstract

We study trade agreements where there is uncertainty about future political-economy value of protection in each country. We assume that the prevailing state of the world is private information of the importing country but it may be verified publicly at a monitoring cost. The optimal trade agreement will be incomplete to save on the cost of implementation of the agreement. We interpret the GATT/WTO agreement as a cap-and-escape arrangement under which governments can increase their tariffs above a pre-determined cap only if they produce evidence that certain states of the world have prevailed. We provide comparative statics results to show how a country's market power affects the use of the two forms of flexibility: the ability to unilaterally change tariffs when tariffs are below the binding and the access to an escape clause. In particular, find that tariff bindings are lower for countries whose preferences are less convergent with those of the world, in the sense that their protectionist bias increases as their private valuation of protection increases. We also show that contingent protection and tariff overhang are substitute and they may not coexist when preferences are sufficiently convergent.

1 Introduction

Some form of escape clause has been a long-standing feature of international trade agreements. The US-Mexico trade agreement of 1942, negotiated as part of the U.S. Reciprocal Trade Agreements Act (RTAA), included a clause indicating that trade concessions could be withdrawn "as a result of unforeseen developments and of trade concessions" that resulted in increased imports and damage to the domestic industry.

*We thank participants at the Midwest International Economics meetings, Dissettle Workshop at ECARES and the Insted Workshop at University of Exeter. Bond thanks the Leitner Center at Yale University for support during work on this project.

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When the RTAA came up for renewal in 1945, Congress exerted pressure on President Truman to include an escape clause in future agreements. This pressure resulted in an executive order by President Truman in 1947 agreements negotiated under the RTAA, and an escape clause with language very similar to that of the US- trade agreement was included as Article XIX of the GATT.[1]

In order for a country to exercise its right under the escape clause, it must conduct an investigation that establishes that there has been an increase in the level of imports and that this increase has caused "serious injury" to the domestic industry. These safeguard actions are an example of contingent protection, which allows for tariff concessions to be withdrawn if an investigation establishes that certain conditions have been met. Antidumping duties, which allow for the imposition of tariffs in the event that imports are being sold at "less than fair value" and are resulting in injury to the domestic industry, represent another example of contingent protection in the WTO.[2]

Contingent protection is not the only form of flexibility that is included in multilateral trade agreements. The fact that countries negotiate bindings, which are maximum tariff rates that can be applied on imports in a particular product classification, means that countries can raise tariffs unilaterally if they are below the binding. The difference between the bound tariff and the applied tariff rate, referred to as "tariff overhang," provides a measure of the amount by which a country can raise its tariff without violating its WTO commitments. Table 1 illustrates the amount of tariff overhang that exists for 66 WTO members that account for 76% of world imports in 2007. Tariff overhang exists in sixty percent of the tariff lines that are bound, and this overhang averages more than 20 percentage points. If unbound tariff lines are included, more than 30% of trade takes place in tariff lines where there is tariff overhang.

Table 1: Tariffs and Trade Summary Statistics

Binding Status	Num. of sector	Share(%)	Import(bil.$)	Share (%)
Applied Tariff below Binding	196,062	65.32	1,760	24.14
Strong Binding (Applied Tariff at Binding)	51,680	17.22	4,410	60.48
Applied Tariff over Binding	8,301	2.76	413	5.66
Unbound	44,136	14.70	709	9.72
Total	300,129	100	7,292	100

Note: Applied tariff data is from 66 WTO members in 2007.

The presence of these forms of flexibility seem to undermine one of the basic objectives of a trade agreement, which is to commit countries to reducing tariff rates. Since

[1] Sykes (2006) provides a good discussion of the history of the safeguard mechanisms.

[2] Under the Safeguards agreement to the WTO, a country imposing safeguards is not required to compensate exporting countries for three years following the imposition of safeguards if there has been an absolute increase in imports. However, compensation must be negotiated if there increase in imports is only relative. The imposition of antidumping duties does not require compensation of affected countries. Both forms of contingent are subject to dispute settlement if the rules are not followed in imposing protection.

increases in tariff rates have a negative effect on trading partners, trade agreements can be welfare improving for all trading countries by reducing the international externalities through mutual tariff reductions. Contingent protection, on the other hand, will be internationally inefficient because they will allow countries to continue to pursue protectionist policies. Thus, many economists have advocated eliminatin

An alternative interpretation of contingent protection and other forms of flexibility, and one that argues for their inclusion in trade agreements, is that they provide a safety valve that allows governments to deal with protectionist pressure. Politically motivated governments are reluctant to make commitments to bind tariffs at low levels because they are concerned about future events that may lead to a demand for a protectionism, and therefore want to have the flexibility to increase tariffs in those cases. In this view, the escape clause provides a safety valve for dealing with such protectionist pressure, and is the price that must be paid for countries to engage in significant trade liberalization.[3]

The purpose of this paper is to develop a model of politically motivated governments that allows for the simultaneous use of both tariff overhang and escape clauses as part of a trade agreement. We work within a simple political-economy trade model in which governments are uncertain about their future preferences over trade policy, so that the country's non-cooperative tariff varies with the magnitude of a political shock. We adopt an incomplete contracting approach in which the magnitude of the political shock, and hence the importer's preferred tariff, is the private information of the country. However, the importing country can make the value of its private information known to the rest of the world by incurring a monitoring cost.

Using this framework, we examine the optimal "cap-and-escape" agreement which has features similar to that of the WTO trade agreement. The cap part of the agreement specifies a tariff binding that allows the country to unilaterally choose any tariff less than or equal to the binding. The escape portion of the agreement allows the country to exceed the binding if the magnitude of the political shock is *verified* to exceeds a threshold level. The magnitude of the shock is verified through a costly state verification process.[4] The threshold level of the shock at which the importer can escape its bound tariff, as well as the magnitude of the tariff that can be imposed when the value of the shock is revealed, are specified as part of the trade agreement. The monitoring that takes place when the importer escapes the tariff binding can be interpreted as the investigative processes required to approve and implement contingent protection under the GATT/WTO agreements. This model yields the result that the escape clause allows for a reduction in tariff bindings in an optimal trade agreement,

[3] Another type of flexibility may be provided through a remedy system in which a party is allowed to breach the contract if it provides appropriate compensations to the affected parties. For a discussion of breach remedies in trade agreements see Posner and Sykes (2011), Beshkar (2010a), and the papers cited in the next footnote.

[4] Other models, including Beshkar (2010b, 2011) Maggi and Staiger (2011), and Park (2011), examine state verification processes that generate only an imperfect, albeit informative, signal of the state of the world. However, these models do not imply a role for tariff overhangs in trade agreements.

and also allows predictions about the relative importance of tariff overhang and the use of escape clause mechanisms.

Kucik and Reinhardt (2009) have found that countries that have antidumping laws in place are more likely to join the WTO and have lower tariff bindings. Since countries that have antidumping laws in place would be expected to have lower costs of undertaking the investigations, their empirical evidence is consistent with our theoretical results on the substitutability between binding overhang and contingent protection as monitoring costs are reduced.

Our analysis is related to several strands of recent work that uses an incomplete contracts approach to modeling trade agreements. Bagwell and Staiger (2005) have shown that the use of a weak tariff binding, which allows a country to choose its tariff at or below the binding, can yield a higher expected welfare than can be obtained by an inflexible tariff rate. Subsequently, Amador and Bagwell (2011) derived conditions under which the use of a tariff binding will be the optimal incentive compatible agreement.[5] Our work extends their analysis by in two ways. First, we use a model with asymmetric countries, which allows us to derive testable implications about the relationship between country characteristics and the optimal amount of tariff overhang. This is of interest because there is substantial variation across countries in the amount of overhang in tariff lines.[6] Second, the existing work on tariff bindings does not allow for the potential interactions between tariff bindings and contingent protection. Nevertheless, contingent protection measures along with weak bindings are two important components of flexibility mechanisms in the WTO agreement.[7] Our work is the first to allow for the use of both bindings and escape clauses as part of optimal agreements, and to relate the usage of the respective types of flexibility to country characteristics.

Our approach is also related to works that examine the effect of transactions costs on the optimal design of contracts. Shavell (2006) and Horn, Maggi, and Staiger (2010) have shown that when writing a contract is costly, it is optimal to craft an incomplete contract in order to save on the ex ante contracting efforts. Our analysis differs in that we emphasize the ex post costs of implementing the contract, which includes the costs of verifying the contingencies that are mentioned in the contract. We find that, due to

[5] The models we consider are closely related to the literature on optimal delegation (e.g., Holmstrom 1984, Melumad and Shibano 1991, and Alonso and Matouschek 2008), which has shown that it may be optimal for a principal to require a privately-informed agent to choose from a restricted set of actions when contingent transfers are not possible.

[6] Almost 2/3 of the tariff lines of 69 WTO members are below their bound rates. The average level of tariff binding overhang, which is the difference between the bound rate and the applied rate, is 25 percentage points or more than a quarter of the bound rates. However, there is no overhang in more than 90% of tariff lines for the US, European Union, China, and Japan. Thus, the usage of tariff overhang as a flexibility mechanism varies widely across countries.

[7] Between 1995 and 2010, a total of 216 safeguard actions and more than 2400 antidumping measures have been notified to the WTO. The importance of contingent protection measures as an element of trade agreements was highlighted in the Doha round of WTO negotiations in which most developing countries demanded the inclusion of a safeguard clause in the agreement as a condition to accept trade liberalization in agricultural sectors.

the costs of implementing a contingent contract, it is optimal to write an incomplete contract that gives discretion to the parties in some contingencies.

Our approach, therefore, provides a distinct and novel rationale for writing an incomplete contract (such as tariff bindings) on the basis of the cost of implementing, rather than writing, the agreement. Implementing a complete agreement, which maps each possible contingency to a set of actions to be taken by the signatories, requires the parties to find a mutual agreement on the nature of the prevailing contingency in each period over the lifetime of the agreement. This requires establishing appropriate procedures for investigation of the contingencies in each member country as well as a dispute settlement process to handle potential disputes over the result of these investigations. Implementation of these procedures are potentially very costly as they involve hiring arbitrators, lawyers, and government representatives.[8] To the best of our knowledge, the previous literature has not explored the impact of the implementation costs on the optimal design of a trade agreement.

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Section 2 presents a general model in which the tariffs of the importing country impose a negative externality on exporting countries and the demand for protection in the importing country. We first examine the case where the tariff binding is the only form of flexibility, and show that the optimal binding will provide flexibility in the form of tariff overhang only if the preferences of the importer are sufficiently aligned with those of the world as a whole. We also relate the level of the optimal binding to the distribution of political shocks and to the degree of convergence/divergence of prefences of the importing country as the political shock increases.

We then characterize the optimal cap-and-escape agreement in which the agreement can use both tariff bindings and contingent protection as flexibility mechanisms. We show that agreements with tariff bindings create an incentive to include an escape

[8] Note that in contrast to implementation costs, the cost of composing a contract is a one-time expense. The fact that the cost of state verification is a recurring cost over the lifetime of the agreement, multiplies the importance of designing a contract that saves on implementation costs.

clause, because both importing country and world welfare will be raised by increasing tariffs for the highest realizations of the political shock. An escape clause in an optimal agreement will specify the first best tariff when monitoring occurs, although the threshold specified in the agreement to allow escape will exceed the level that is preferred by the importing country. We also establish that contingent protection is a substitute for tariff overhang, because it will result in a reduction in the tariff binding in the optimal trade agreement.

Section 3 provides examples of trade models that generate preferences consistent with the assumptions of the general model. We show that in a perfectly competitive model where the externality is transmitted through the terms of trade, the magnitude of the externality is larger the greater is the market power of the importing country. Market power is increasing in the relative size of the importing country and the degree of its comparative disadvantage.

2 A General Model with Uncertainty About Value of Protection

2.1 Preferences

We begin our analysis with a very general model of sectoral trade protection that emphasizes two features. The first is that home country tariff policy has negative spillovers on foreign exporters, so unilateral tariff setting will represent a prisoner's dilemma. This externality can be due to a terms of trade effects in perfectly competitive models as well as in other The second feature is that the importing country is subject to random shocks that affect its valuation on protection. The countries know the distribution of these shocks prior to signing a trade agreement, but realizations occur after the agreement has been put in place and are the private information of the importing country. .

We represent the preferences of the home country over tariff rates in a representative import-competing sector by a sectoral welfare function $V(t, \theta)$, where t is the tariff imposed on imports in that sector and θ is a random variable reflecting political shocks that affect the demand for protection. The random variable θ has a pdf $f(\theta)$ that has compact support $\Theta = [0, \theta^{\max}]$. We make the following assumptions on the importing country's preferences

Assumption 1 *$V(t, \theta)$ is strictly quasi-concave in t on $[0, t^P]$, with $V_{t\theta} > 0$. The importer's optimal (non-cooperative) tariff on the importable, $t^N(\theta)$ is uniquely determined with $\frac{dt^N(\theta)}{d\theta} = -\frac{V_{t\theta}}{V_{tt}} > 0$.*

The Higher values of θ are associated with a greater marginal benefit from protection when the tariff is below the prohibitive level. With these assumptions, the importing

country's optimal (Nash) tariff on the importable good as $t^N(\theta)$, where $\frac{dt^N(\theta)}{d\theta} = -\frac{V_{t\theta}}{V_{tt}} > 0$.

The welfare of the foreign country in this sector is denoted $V^*(t)$, where V^* is decreasing in t for $t < t^P$.Assuming that transfers between countries are possible ex ante, we can view a trade agreement as maximizing the sum of home and foreign welfare in each sector. World welfare for a representative home importable sector is

$$W(t, \theta) = V(t, \theta) + V^*(t), \tag{1}$$

We assume that world welfare function has the following properties:

Assumption 2 *$W(t, \theta)$ is strictly quasi-concave in t. The state contingent tariff that maximizes world welfare, $t^E(\theta) = \arg\max W(t, \theta)$, will be increasing in θ under our assumptions since $\frac{dt^E}{d\theta} = -\frac{W_{t\theta}}{W_{tt}} = -\frac{V_{t\theta}}{W_{tt}} > 0$.*

The gains from reciprocal trade liberalization are reflected in the fact that our assumptions ensure that $t^N(\theta) \geq t^E(\theta)$, with strict inequality for $t^N(\theta) < t^P$.

2.2 The Optimal Tariff Binding

We now turn to the analysis of trade agreements in settings where the magnitude of the political pressure is the private information of the importing country. We will assume that the distribution of the political shock in each country is common knowledge, but the realization is observed only by the importing country. We will also assume that transfers between countries are not possible ex post, so that the problem can be treated as a delegation problem in which the uninformed principal (the WTO) specifies a set of actions (tariff levels) from which the informed agent (the importing country) can choose. This problem can be formulated as one in which the tariff schedule under the trade agreement, $t(\theta)$, must satisfy the incentive compatibility constraint,

$$V(t(\theta), \theta) - V(t(r), \theta) \geq 0 \text{ for all } r, \theta \in \Theta \tag{2}$$

where r is the importing country's report of the state. The tariff schedule $t^E(\theta)$ is clearly not incentive compatible when θ is private information, because a country receiving a shock θ would have an incentive to report the value as $\theta' = t^{E^{-1}}(t^N(\theta)) > \theta$.

In this section we examine the optimal tariff policy when the schedule takes the form of a tariff binding, t^B, which allows the importing country to choose any tariff $t \leq t^B$ without violating the agreement. In the presence of a tariff binding, the importer's welfare is maximized by choosing an applied tariff of $\min[t^N(\theta), t^B]$ when the political shock is θ. The tariff binding can be represented by the incentive compatible tariff schedule,

$$t(\theta) = \begin{cases} t^B, & \text{if } \theta \geq \theta^B(t^B) \equiv \max\{\underline{\theta}, t^{N^{-1}}(t^B)\}, \\ t^N(\theta), & \text{if } \theta < \theta^B(t^B). \end{cases} \tag{3}$$

We will refer to the set $B = [\theta^B(t^B), \bar{\theta}]$ as the binding region, as it is the set of states of the world at which importing country policy is constrained by t^B. Our focus in this section is on how sectoral characteristics affect the amount of flexibility provided to the importing country in a trade agreement. If $t^B > t^N(\underline{\theta})$, there will exist states of the world for which the tariff is strictly less than the binding, which we refer to as an agreement with tariff overhang. If $t^B \leq t^N(\underline{\theta})$, the importing country has no flexibility because its tariff is always at the binding.

We assume that transfers between countries are available ex ante, so that the objective for the trade agreement is to maximize expected world welfare subject to (3). This optimization problem can be expressed as

$$\max_{t^B} E\left[W\right] = \int_{\underline{\theta}}^{\theta^B(t^B)} W(t^N(\theta), \theta) f(\theta) d\theta + \int_{\theta^B(t^B)}^{\bar{\theta}} W(t^B, \theta) f(\theta) d\theta,$$

where,

$$\theta^B(t^B) = \max\{\underline{\theta}, t^{N^{-1}}(t^B)\}. \tag{4}$$

The necessary condition for a maximum at an interior solution is

$$\int_{\theta^B}^{\bar{\theta}} W_t(t^N(\theta^B), \theta) f(\theta) d\theta = 0 \tag{5}$$

Using (1), this condition equates the cost to the exporting country of raising the tariff, $-V_t^*(t)$, to the expected benefit to the importing country of raising the tariff, $\frac{\int_{\theta^B}^{\bar{\theta}} V_t(t^N(\theta),\theta) f(\theta) d\theta}{1-F(\theta^B)}$.

Equation (5) will also be sufficient for a maximum if

$$\Lambda^B \equiv \left(\int_{\theta^B}^{\bar{\theta}} W_{tt}(t^N(\theta^B), \theta) f(\theta) d\theta \right) - \frac{W_t(t^N(\theta^B), \theta^B) f(\theta^B)}{\frac{\partial t^N(\theta^B)}{\partial \theta}} < 0 \tag{6}$$

The first term is the effect of an increase in the binding on the on expected marginal benefit of protection on world welfare, holding the binding region B constant. This term will be negative due to the concavity of the world welfare function. The second term captures the fact that an increase in the binding will reduce the binding region, which will raise the marginal benefit of protection because $W_t(t^N(\theta^B), \theta^B) < 0$. Thus, sufficiency requires that the first term be larger than the second.

Figure 1 illustrates the determination of the optimal binding. Let

$$\bar{t}^E(\theta) = \arg\max \int_{\theta}^{\bar{\theta}} W(t, z) \left(\frac{f(z)}{1 - F(\theta)} \right) dz \tag{7}$$

denote the tariff that maximizes world welfare conditional on a realization exceeding θ. This curve will be upward sloping, and must satisfy $t^E(\underline{\theta}) < \bar{t}^E(\theta) < t^E(\bar{\theta})$ for $\theta < \bar{\theta}$ as

illustrated in Figure 1. The necessary condition for the optimal binding at an interior solution can be written as

$$t^N(\theta^B) = \bar{t}^E(\theta^B),$$

as illustrated by the intersection of the respective schedules in Figure 1. The sufficient condition for a maximum requires that the $\bar{t}^E(\theta^B, \bar{\theta})$ locus be steeper at an intersection.

There will be no overhang under the optimal binding if $t^N(\theta) > \bar{t}^E(\theta)$ for all θ in Θ. The optimal agreement schedule is given by $\min[t^N(\theta), t^B]$,illustrated by the thick locus in Figure 1. Note that in the region where the tariff is at the binding, there will be a value $\tilde{\theta} = t^{N^{-1}}(t^B)$ such that $t^E(\theta) < (>) t^B$ for $\theta < (>)\ \tilde{\theta}$.

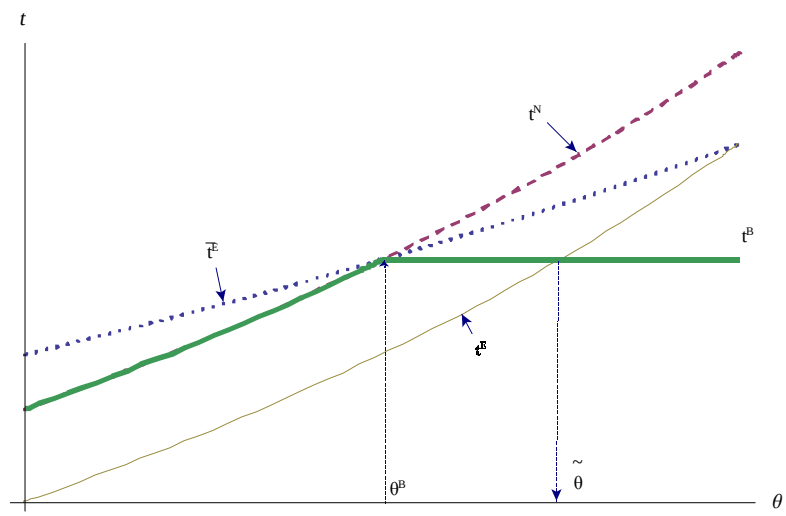

Figure 1: The optimal binding with overhang

In the case of trade agreements, one factor in determining the degree to which preferences are aligned is the magnitude of the spillover of home tariffs to the rest of the world. We can illustrate the role of the magnitude of the spillover on the optimal binding by considering the effect of an increase in the value of protection to the importer combined with an equal cost of protection to the exporter, so that the world welfare function remains unchanged.

Proposition 1 *Let the preferences of the home country be given by $V(t, \theta) + \mu t$ and foreign preferences by $V^*(t) - \mu t$. At an interior solution, an increase in μ reduces both θ^B and t^B. At a corner solution with no overhang, an increase in μ has no effect on the binding.*

An increase in μ raises the importing country's non-cooperative tariff, while leaving the efficient tariff unaffected. Referring to Figure 1, an increase in μ shifts the $t^N(\theta)$ locus upward. The new optimum will occur at a lower value of both θ^B and t^B. It might seem surprising that an increase in the non-cooperative tariff reduces the optimal binding, since the world welfare maximizing tariff is unaffected by μ. This result arises

because the threshold value of the shock at which the binding applies, (4) is decreasing in μ. An increase in μ will expand the range of states for which $t^E(\theta) < t^B$, which reduces the value of flexibility and leads to a lower optimal binding.

The variation in importer preferences considered in 1 raises the value of protection by the same amount for all values of θ. However, the analysis shows that the comparative static result depends only on the change in the degree of protectionism of the importer in the neighborhood of the binding, and not on the average degree of protectionism. The following example provides an example of a change in importer preferences that holds the average degree of protectionism constant, while making the importer more protectionist in the neighborhood of the binding.

Corollary 3 *Consider a rotation in importer preferences around $\theta_0 \in [\theta^B, \bar{\theta}]$ such that $t^N(\theta)$ increases for $\theta < \theta_0$ and decreases for $\theta > \theta_0$, while keeping $\int_{\theta^B}^{\bar{\theta}} t^N(\theta) f(\theta) d\theta$ and $W(t, \theta)$ constant. This variation will reduce the optimal binding.*

The variation in preferences considered in the corollary "rotates" the non-cooperative schedule of the importer so that it becomes less divergent from $t^E(\theta)$ as θ increases, while maintaining the average degree of protectionism above the binding. This change reduces the optimal degree of flexibility in the agreement. This suggests that the binding will tend to be lower when the preferences of the importer and the world are more convergent as θ increases.

The optimal binding will also be affected by the distribution of political shocks. Referring to Figure 1, it can be seen that an increase in political pressure that raises $\bar{t}^E(\theta)$ will result in an increase in the optimal binding at an interior solution. In order for a change in the distribution of political shocks to raise $\bar{t}^E(\theta)$, it must put relatively greater weight on larger on larger values of the political shock. To see this, let $f(\theta, \xi)$ be a parameterized family of densities with common support $[\underline{\theta}, \bar{\theta}]$ that has the property of log supermodularity. The property of his condtional distribution has the property of supermodularity requires that for $\theta_2 > \theta_1$ and $\xi_2 > \xi_1$ we have $\frac{f(\theta_2,\xi_2)}{f(\theta_2,\xi_1)} > \frac{f(\theta_1,\xi_2)}{f(\theta_1,\xi_1)}$. it is clear that the conditional densities, $\frac{f(\theta,\xi)}{1-F(\theta^B,\xi)}$, will also have the property of log supermodularity. Since $W_{t\theta} > 0$, it follows from Theorem 3 in Athey (2002) that $\bar{t}^E(\theta)$ will be non-decreasing in ξ. An increase in the tariff has greater value when θ is large, so the value of the binding will be higher for distributions that assign relatively greater probability to larger values of θ.

These results establish:

Proposition 2 *Consider a family of density functions $f(\theta, \xi)$ with common support Θ that is log supermodular. The optimal tariff binding will be non-decreasing in ξ at any interior solution.*

Note that while the densities $f(\theta, \xi)$ results in a at a ranking of first order stochastic dominance of the distributions, a ranking of distributions based on first order stochastic dominance by itself is not suffficient to establish the result.

2.3 Cap and Escape Agreement

We now consider the possibility that by incurring a monitoring cost of c, the importing country is able to reveal the true value of θ to the rest of the world. We can then specify a trade agreement as consisting of two regions: a monitoring region ($M \sqsubseteq \Theta$) and a non-monitoring region (M^C). A country reporting a state in the non-monitoring region is assumed to be subject to a tariff binding, as in the previous section. If the importing country incurs the monitoring cost and a state $\theta \in M$ is verified, then it receives the tariff $t^M(\theta)$ that is specified as part of the agreement. If the importer incurs the monitoring cost and the state is revealed to be outside the monitoring region, then it is subject to the binding. The monitoring region corresponds to the escape clause of a trade agreement, and represents the conditions that must be met in order for the country to be able to violate the binding.

The importer's announcement of a state in the monitoring region can be thought of as the initiation of an investigation and the incurring of the resource costs required to establish that the conditions required for violating the binding are met. While political pressure may not be directly observable, factors such as losses in profits and rising unemployment in the industry that are highly correlated with political pressure can be verified by an investigation. The factors required to establish substantial injury to justify a safeguard action under WTO rules are likely to be correlated with political pressure.

Once the true state is revealed, the importer can impose the tariff $t^M(\theta)$. In order for $t^M(\theta)$ to be incentive compatible, the importing country must prefer the payoff it receives from undergoing monitoring to any tariff that it could choose in the non-monitoring region, namely

$$V(t(\theta), \theta) - V(t(r), \theta) \geq c \text{ for all } \theta \in M, r \in M^C. \tag{8}$$

For $\theta \in M^C$, there is no incentive to choose a report in the monitoring region because the assigned tariff will be the same as if no monitoring had occurred but the cost of monitoring will be incurred.

The potential for such an agreement to improve on the agreement with a tariff binding alone can be seen from the necessary condition for the optimal binding, (**??**), which shows that the binding will exceed the efficient tariff at θ^B and will be less than the efficient tariff in the neighborhood of $\overline{\theta}$. For values of θ sufficiently high, an increase in the applied tariff would benefit both the importing country and the world as a whole. Offering a high tariff in the event that a high value of the political shock is verified has the potential to be both welfare improving and incentive compatible if c is not too large. World welfare can also be improved if tariffs are reduced for low realizations of θ, since the efficient tariff is below that specified under the binding. However, monitoring would not be incentive compatible in this case because the importer prefers the tariff offered under the binding to the efficient tariff.

A cap-and-escape agreement can be characterized by a tariff binding, t^B, a threshold value, θ^M, for the monitoring region, $M = [\theta^M, \overline{\theta}]$, and a tariff schedule for the

monitoring region, $t^M(\theta)$. We begin our analysis by establishing the following Lemma, which characterizes the optimal escape rule $\{\theta^M, t^M(\theta)\}$ given t^B.

Lemma 1 *Suppose that for a given t^B, there exists a $\hat{\theta} < \overline{\theta}$ such that $W(t^E(\hat{\theta}), \hat{\theta}) - c = W(t^B, \hat{\theta})$. Then given t^B, the optimal escape rule is given by $t^M(\theta) = t^E(\theta)$ for $\theta \in M = [\hat{\theta}, \overline{\theta}]$.*

Proof. For $\theta \leq t^{N^{-1}}(t^B)$, the only incentive compatible tariff schedule is given by $t^N(\theta)$ and, hence, monitoring in this region is not optimal. For $\theta > t^{N^{-1}}(t^B)$, monitoring will be incentive compatible iff $V(t^M(\theta), \theta) - V(t^B, \theta) \geq c$. Since $W(t^M(\theta), \theta) - W(t^B, \theta) = \big(V(t^M(\theta), \theta) - V(t^B, \theta)\big) + \big(V^*(t^M(\theta)) - V^*(t^B)\big)$ and $V^*(t)$ is decreasing in t, any agreement with $t > t^B$ that raises world welfare is also incentive compatible. Therefore, the monitoring region should consist of all θ such that world welfare can be raised by monitoring. World welfare cannot be improved by monitoring for $\theta \in [t^{E^{-1}}(t^B), \hat{\theta}]$, because $\frac{\partial W(t^E(\theta),\theta) - W(t^B,\theta)}{\partial \theta} > 0$ for $t^E(\theta) > t^B$. For $\theta \geq \hat{\theta}$, world welfare is maximized at $t^E(\theta)$. Therefore, since $t^E(\theta)$ is also incentive compatible for $\theta \geq \hat{\theta}$, the optimal escape rule is $t^M(\theta) = t^E(\theta)$ for $\theta \geq \hat{\theta}$. ■

Figure 2 illustrates the form of the tariff schedule under a cap and escape agreement as given by Lemma 1. For $\theta < \theta^M$, the importer's tariff is determined by the tariff binding. For $\theta \geq \theta^M$, the importer incurs the cost of monitoring and receives the efficient tariff, with the boundary of the monitoring region chosen to make world welfare equal under monitoring and the binding. Lemma 1 shows that monitoring occurs less frequently than would be desired by the importing country, because θ^M exceeds the value of θ at which the importing country is indifferent between t^B and incurring the monitoring costs to obtain the efficient tariff.[9] As a result, the importing country's incentive constraint (8) will be slack for any realization of θ at which world welfare would be improved by monitoring and the efficient tariff can be imposed in the monitoring region.

Note that this result would continue to hold if part of the monitoring costs were paid by the rest of the world, since this would only have the effect of further relaxing the importer's reporting constraint.

[9] This result is reminiscent of the 'serious' injury condition under the WTO agreements on safeguards and antidumping, which precludes the use of contingent protection measures in cases where the magnitude of alleged injury to domestic industries is not sufficiently great.

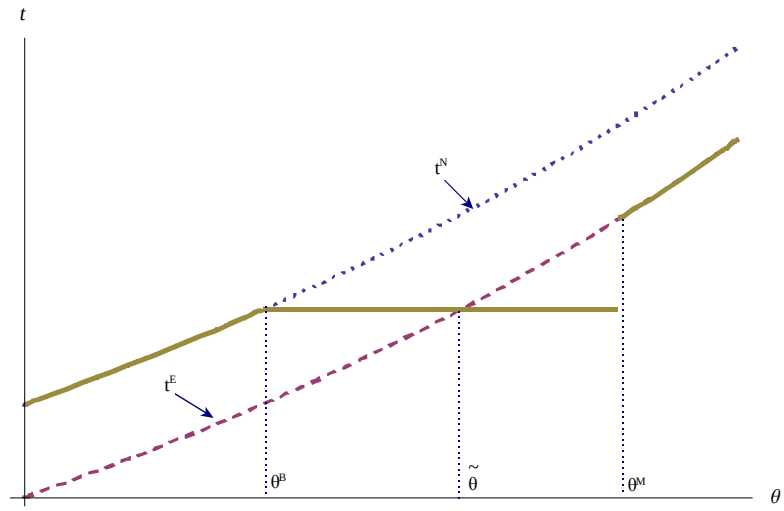

Figure 2: The Cap and Escape Tariff Schedule (Solid Line)

The lower boundary of the monitoring region is the solution to

$$W(t^E(\theta^M), \theta^M) - W(t^B, \theta^M) - c = 0. \tag{9}$$

Since $W_\theta(t^E(\theta^M), \theta^M) - W_\theta(t^B, \theta^M) > 0$, we can invert (9) to obtain $\theta^M(t^B, c)$ and

$$\frac{\partial \theta^M}{\partial t^B} = \frac{W_t(t^B, \theta^M)}{W_\theta(t^E(\theta^M), \theta^M) - W_\theta(t^B, \theta^M)} > 0, \quad \frac{\partial \theta^M}{\partial c} = \frac{1}{W_\theta(t^E(\theta^M), \theta^M) - W_\theta(t^B, \theta^M)} > 0, \tag{10}$$

A higher bound tariff or an increase in the cost of montoring will reduce the gain to monitoring, which raises the boundary of the monitoring region.

Given the characterization of the monitoring region from Lemma 1, the optimal trade agreement is obtained by choosing t^B to maximize

$$EW(t^B) = \int_{\underline{\theta}}^{\theta^B(t^B)} W(t^N(\theta), \theta) f(\theta) d\theta + \int_{\theta^B(t^B)}^{\theta^M(t^B, c)} W(t^B, \theta) f(\theta) d\theta + \int_{\theta^M(t^B, c)}^{\overline{\theta}} \left(W(t^E(\theta), \theta) - c \right) f(\theta) d\theta.$$

The necessary condition for this case is

$$\int_{\theta^B(t^B)}^{\theta^M(t^B, c)} W_t(t^B, \theta) f(\theta) d\theta = 0 \tag{11}$$

Letting $\hat{t}^B$ be the solution to the necessary condition in (5) for the case without monitoring, we have $\int_{\theta^B(\hat{t}^B)}^{\theta^M(\hat{t}^B, c)} W_t(\hat{t}^B, \theta) f(\theta) d\theta < 0$ if $\theta^M(\hat{t}^B, c) < \bar{\theta}$. The presence of a monitoring region reduces the value of raising the binding, because monitoring replaces the binding for the highest values of θ.

2.4 Substitutability Between Overhang and Escape

The fact that the presence of overhang reduces the value of raising the binding for a given t^B in (11) suggests that the possibility of an escape clause should result in a reduction in the optimal binding. We can formalize this by noting that $\frac{\partial^2 EW(t^B)}{\partial t^B \partial c} = W_t(t^B, \theta^M) f(\theta^M) \left(\frac{\partial \theta^M}{\partial c}\right) \geq 0$, with strict inequality holding at an interior solution. The benefit of raising the binding is non-decreasing in c, which means that the expected world welfare function is supermodular in (t^B, c). We can then apply the monotone comparative statics results of Milgrom and Shannon and Edlin and Shannon to obtain the following result:

Proposition 3 *The optimal tariff binding is non-decreasing in c, and will be strictly increasing at an interior solution. The size of the monitoring regions is non-increasing in c, and will be strictly decreasing at an interior solution.*

Supermodularity of the expected welfare world welfare function yields the comparative static results for the binding. This result is useful because it establishes that the substitutability between binding and monitoring does not require that the expected welfare function be concave in t^B.

A natural question to ask is the degree of substitutability between overhang and monitoring. Will the optimal agreement involve the use of both monitoring and overhang, or is the degree of substitutability so high that the optimal agreement uses only one form of flexibiltiy? To address this question, we examine the second order condition for the optimization problem with monitoring:

$$\Lambda^M \equiv \int_{\theta^B(t^B)}^{\theta^M(t^B,c)} W_{tt}(t^B, \theta) f(\theta) d\theta - \frac{W_t(t^B, \theta^B) f(\theta^B)}{\frac{\partial t^N(\theta^B)}{\partial \theta}} - W_t(t^B, \theta^M) f(\theta^M) \frac{\partial \theta^M}{\partial t^B} \quad (12)$$

The interpretation of (12) is similar to that in (6). The first term is the the impact on the expected benefit of protection of an increase in the binding for a fixed range or shocks, which will be negative . The remaining terms capture the effect of an increase in t^B on the size of the binding region. In this case an increase in t^B raise both the lower and upper boundaries of the binding region, both of which terms are positive. In order for (11) to be a maximum, the first effect must dominate the last two. .One case in which $\Lambda^M < 0$ will where $\frac{f(\theta^B)}{F(\theta^M)-F(\theta^B)}$ and $\frac{f(\theta^M)}{F(\theta^M)-F(\theta^B)}$ are sufficiently small, so that the first term must dominate the remaining two. This is a case in which political shocks are concentrated in the interior of the binding region.

A second factor that plays a role in the substitutability between overhang and escape is the extent to which the preferences of the importer and the world converge as the value of θ increases. We define

$$\Delta(\theta) = t^N(\theta) - t^E(\theta)$$

to be the difference between the Nash tariff and the efficient tariff, and refer to preferences as *locally convergent* at θ if $\Delta'(\theta) < 0$. The following example with quadratic preferences, which results in a constant value of $\Delta'(\theta)$ for all θ, shows that convergence of preferences is a sufficient condition to rule out the use of both overhang and escape in an optimal agreement when θ is uniformly distributed.

Proposition 4 *Assume preference functions are given by* $W(t,\theta) = A(\theta) + a^W t - \frac{b^W}{2}t^2 + \theta t$ *and* $V(t,\theta) = A(\theta) + at - \frac{b}{2}t^2 + \theta t$.

(i) $V_{t\theta} = W_{t\theta} = 1$

(ii) $V_{tt}(t,\theta) = -b < 0$ *and* $W_{tt}(t,\theta) = -b^W < 0$

If θ *is uniformly distributed, there is no interior solution for a binding when* $b > b^W$.

With the quadratic preferences in Proposition 4, $\Delta'(\theta) = \frac{1}{b} - \frac{1}{b^W}$, so $b > b^W$ ensures convergence of preferences. This result shows that overhang and escape are extremely good substitutes under the assumptions of Proposition 4. This result is clearly related to Corollary 3, which showed that a change in the importer preferences that made them more convergent to those of the world resulted in a reduction in the binding. Together, these results suggest that overhang is less valuable and more susbtitutable for an escape clause when preferences are more convergent.

Since the externality from trade agreements Proposition 4 shows that the downward pressure on the binding resulting from in a similar vein, since it indicates that overhang will not be use

2.5 Escape and the Alignment of Preferences

One important difference between overhang and escape clauses as a means of providing flexibility is that monitoring does not require an alignment of preferences between the importing country and the rest of the world. Consider the case in which $t^N(\underline{\theta}) > \bar{t}^E(\theta)$ for all θ in Θ, so that the solution to (5) will be $t^B = \bar{t}^E(\underline{\theta})$ and will provide no flexibility to the home country. However, the agreement will include monitoring if $c < W(t^E(\bar{\theta}),\bar{\theta}) - W(\bar{t}^E(\underline{\theta}),\bar{\theta})$ In particular, note that changes in preferences of the type considered in Proposition 1 will have no effect on the optimal monitoring regions in this case. The function $\theta^M(t^B, c)$ does not depend on foreign preferences because the importing country's incentive constraint does not bind, and t^B is independent of importer preferences when $t^B < t^N(\underline{\theta})$.

We can also illustrate the effect of alignment of preferences on the use of bindings in the neighborhood of an interior solution by considering the exercise from Proposition 1:

Proposition 5 *Let the preferences of the home country be given by* $V(t,\theta) + \mu t$ *and foreign preferences by* $V^*(t) - \mu t$. *At an interior solution, an increase in* μ *reduces* θ^B, θ^M *and* t^B.

A larger divergence in preferences between the importing country and the world as a whole results in a substitution of bindings for monitoring in the optimal trade agreement. The argument is essentially the same as for Proposition 1 because the only effect of a change in μ, given and interior solution t^B, is to reduce θ^B. This results in a reduction in t^B, which causes an expansion of the monitoring region by (10).

3 Examples

We now examine two specific models to illustrate the factors that affect the form of the optimal trade agreement. The first is a competitive model in which externalities arise through the terms of trade. The second is a monopolistic competition model in which the externality arises from the impact on market shares of the exporting firms.

3.1 Terms of Trade Externalities

We first consider a two-country, many good trade model with perfectly competitive industries based on Bond and Park (2002). In this model the spillover of trade policy operates through the terms of trade. Our objective is to show that this model generates sectoral objective functions that have the properties discussed in the previous sections, and to illustrate how the degree of alignment of preferences is related to the relative size of the countries and the degree of comparative advantage.

The home country demand for good i is given by $d_i = \lambda(1 - p_i)$ for $i = 1, .., I$, where p_i is the price of good i and $\lambda \in (0, 1)$ is the relative size of the home country. Home supply is $x_i = \lambda\beta_i p_i$, which results in an autarky price of $1/(1 + \beta_i)$. Foreign demand is supply is $d_i^* = (1 - \lambda)(1 - p_i^*)$ and foreign supply is $x_i^* = (1 - \lambda)\beta_i^* p_1^*$, so the foreign country has comparative advantage in sector i if $\beta_i < \beta_i^*$. This model as a partial equilibrium flavor, which allows us to advantage can be derived from a general-equilibrium model with a good 0 that absorbs all income effects. [10]

In light of the separability of markets, the effect of home tariff policy in one sector is independent of tariffs in other sectors. Therefore, we consider a representative home importable where we choose $\beta \equiv 1$ and let $\beta^* > 1$ represent the degree of foreign country comparative advantage. Letting t be the ad valorem tariff imposed by the home country on imports, prices in the respective markets are linked by $p = p^*(1 + t)$.

[10] Let the home country consist of a measure of N identical households with each household having a utility function $U = \sum_{i=1}^{n} d_i(1 - .5d_i) + d_0$. Households have an endowment of labor that can be allocated to production of the three goods. Letting l_i denote the quantity of labor allocated to good i, the production functions are $x_0 = l_0$, $x_1 = (2l_1)^{.5}$ and $x_2 = (2\beta l_2)^{.5}$. Similarly, the foreign country is assumed to have N^* households with the same preferences and production functions $x_0^* = l_0^*$, $x_1^* = (2\beta l_1^*)^{.5}$, and $x_2^* = (2l_2^*)^{.5}$. Choosing good 0 as numeraire and letting $\lambda = N/(N + N^*)$, this structure yields the demand and supply functions in the text if the supply of labor is sufficiently large that good 0 is always produced.

The market-clearing prices with trade,

$$p^*(t) = \frac{1}{2\lambda(1+t)+(1+\beta)(1-\lambda)}, \quad p(t) = \frac{1+t}{2\lambda(1+t)+(1+\beta)(1-\lambda)}. \tag{13}$$

The home country tariff improves the terms of trade of the home country. The magnitude of the terms of trade externality depends on the relative size of the countries, with $dp^*/dt \to 0$ as $\lambda \to 0$ and $dp^*/dt \to -1$ as $\lambda \to 1$. The prohibitive tariff will be $t^{pro} = \frac{\beta-1}{2}$.

We assume that the government's preference over tariffs can be described by a 'political' welfare function, where the government puts a weight of $1+\theta$ on the welfare of producers in the import-competing sector and a weight of 1 on the welfare of all other agents. We assume that $\theta \geq 0$, and interpret θ as the political pressure that is exerted by the import-competing sector on the government. The political welfare function may be written as

$$V(t,\theta) = S(t) + (1+\theta)\pi(t) + tp^*(t)m(t), \tag{14}$$

where, consumer surplus is given by $S(t) = \lambda(1-p(t))^2/2$, producer surplus by $\pi(t) = \lambda p(t)^2/2$, and tariff revenue by $tp^*m(t) = tp^*(t)\lambda(1-2p(t))$. Increases in t have a favorable effect on political welfare by improving the terms of trade and transferring income to domestic producers of import-competing goods (when $\theta > 0$). However, increases in t also reduce trade volume, which is welfare reducing when the domestic price exceeds the world price for importables. As a result of these trade-offs, home country welfare is strictly quasi-concave in t for $t \in [0, t^{pro}]$.[11]

The non-cooperative tariff of the home country has the following properties

Lemma 2 *For $\theta < \theta^{\max}$ and $\lambda \in (0,1)$, the non-cooperative equilibrium tariffs are given by*

$$t^N(\theta) = \frac{\theta(1+\beta)+2(\beta-1)\lambda}{(2-\theta)(1+\beta)+4\lambda}. \tag{15}$$

The non-cooperative tariffs have the properties $t^N_\theta(\theta) > 0$, $t^N_\beta(\theta) > 0$, $t^N_\lambda(\theta) > 0$.

The sectoral welfare function for the importer satisfies Assumption 1. We let $\theta^{\max} \equiv 2(\beta-1)/(1+\beta) < 2$ denote the value of the political pressure, θ, at which the home country's non-cooperative tariff eliminates trade, $t^N(\theta^{\max}, \lambda) = t^{pro}$. The non-cooperative tariff is increasing in β and λ, which reflects the effect of these parameters on the importer's market power.

For the foreign country, welfare is the sum of consumer surplus and firm profits, $\sqrt{B}$

$$V^*(t) = S^*(t) + \pi^*(t), \tag{16}$$

[11] The impact of t on home welfare is $V_t = \frac{\lambda(1+\beta)(1-\lambda)(\theta(1+\beta)+2(\beta-1)\lambda-t((2-\theta)(1+\beta)+4\lambda))}{(\beta(1-\lambda)+\lambda(1+2t)+1)^3}$. Political welfare is increasing for $t < t^N(\theta)$ and decreasing for $t > t^N(\theta)$, so V is quasi-concave in t for $t \in [0, t^{pro}]$.

where $S^*(t) = (1-\lambda)(1-p^*(t,\lambda))^2/2$ and $\pi^*(t) = (1-\lambda)\beta^* p^*(t)^2/2$. Foreign welfare is decreasing and convex in t. An increase in the home tariff worsens the terms of trade for the foreign country, which reduces foreign welfare. The convexity of foreign welfare arises because the adverse terms of trade effect is proportional to the volume of foreign exports, and the volume of exports declines with increases in t. Defining world welfare as in (**??**), $W(t,\theta)$ is quasiconcave in t for $\theta \in [0, \theta^{\max}]$, and achieves a maximum at [12]

$$t^E(\theta) = \frac{\theta}{2-\theta}. \tag{17}$$

World welfare satisfies Assumption 2. Since world welfare is unaffected by changes in β and λ, it follows immediately from Proposition 1 that the tariff binding will be inversely related to these paramters at an interior solution.

Using the fact that $-\frac{W_t(t^B,1)}{\pi_t(t^B)} = \frac{1-t^B}{1+t^B}$ with this welfare function, the necessary condition for an optimal binding (5) can be written as

$$\frac{1-t^B}{1+t^B} = E[\theta|\theta \geq \theta^B = \max\{\underline{\theta}.\frac{2t(1+\beta)+4\lambda(t-t^{pro})}{(1+t)(1+\beta)}] \tag{18}$$

The left hand side of this equation is the marginal welfare cost per dollar of profit generated by raising the binding, and the right hand side is the expected political gain obtained from an additional dollar of profit over the region where the tariff is at the binding. Higher bindings raise the marginal welfare cost, but also raise the expected value of the raising idnig

We can obtain a closed form solution for the optimal binding if we assume that θ the shocks have a power function distribution given by

$$f(\theta,\gamma) = \frac{\gamma\left(\overline{\theta}-\theta\right)^{\gamma-1}}{\left(\overline{\theta}-\underline{\theta}\right)^{\gamma}} \text{ for } \theta \in [\underline{\theta}, \overline{\theta}] \tag{19}$$

The power function yields a uniform distribution for $\gamma = 1$, with $f'(\theta) > 0$ (< 0) for $\gamma > 1$ (< 1). This distribution has the property that if $\gamma_2 > \gamma_1$, $f(\theta,\gamma_2)$ is stochastically smaller according to the monotone likelihood ranking because it puts less weight on large value of θ. The mean of θ in this case is $\frac{\overline{\theta}+\gamma\underline{\theta}}{1+\gamma}$, so larger values of γ result in a lower expected value for the political shock.

Substituting (19) into the necessary condition for the optimal binding yields the following result:

Proposition 6 *Assume that $\overline{\theta} < \theta^{\max} \equiv 2\left(\frac{\beta-1}{\beta+1}\right)$ and the political pressure parameter has a power-function distribution as given by (19). The optimal binding will have the following properties:*

[12] Derivative of the world welfare with respect to tariff is $W_t = \frac{\lambda(1+\beta)(1-\lambda)(\theta-t(2-\theta))}{(\beta(1-\lambda)+\lambda(1+2t)+1)^3}$. As is clear from this expression, world welfare is increasing for $t < \frac{\theta}{2-\theta}$ and decreasing for $t > \frac{\theta}{2-\theta}$. Therefore, W is quasiconcave and $t^E(\theta)$ is the jointly optimal tariff.

(i) If $\lambda < \tilde{\lambda} \equiv \frac{\overline{\theta}-\underline{\theta}}{(1+\gamma)\theta^{\max}-\overline{\theta}-\gamma\underline{\theta}} < \frac{1}{\gamma}$*, there will be a unique interior optimal binding* $t^B = \frac{\overline{\theta}-\gamma\lambda\theta^{\max}}{2-\overline{\theta}-4\gamma\lambda/(1+\beta)}$*. This tariff binding is the optimal among all incentive compatible trade agreements.*

(ii) If $\lambda \geq \tilde{\lambda}$*, the optimal binding agreement will involve a tariff binding* $t^B = \frac{\overline{\theta}+\underline{\theta}\gamma}{2(\gamma+1)-\left(\overline{\theta}+\underline{\theta}\gamma\right)}$ *with no overhang.*

The preferences of the importer are sufficiently aligned with those of the world that the efficient agreement involves tariff overhang if $\lambda < \tilde{\lambda}$. The optimal binding is decreasing in γ, reflecting the result of Proposition 2. It is shown in the Appendix that an interior solution satisfies the sufficient conditions of Amador and Bagwell for interval delegation to be optimal, so that in this case the tariff binding is optimal among all incentive compatible trade agreements.

The corollary to Proposition 1 shows that a rotation of the importer's non-cooperative tariff schedule that increases the gap between the importer's otpimal tariff and the efficient tariff at high values of θ will raise the optimal binding. This result suggests a role that it is useful to identify the extent to which the preferences of the importer are converging or diverging from those of the world as a whole as θ increases. We say that the preferences of the importer and the world are '*locally convergent*' at θ if $\Delta'(\theta) < 0$.

Lemma 3 (Convergence of Preferences) *Defining* $\tilde{\beta}(\theta) \equiv \frac{2+\theta+4\sqrt{1+\lambda}}{2-\theta}$,

(i) The preferences of the importer and the world are locally convergent at θ *(i.e.,* $\Delta'(\theta) < 0$*) iff* $\beta < \tilde{\beta}(\theta)$*. For* $\beta < \tilde{\beta}(0) = 1 + 2\sqrt{1+\lambda}$*, preferences are globally convergent.*

(ii) Consider $d\beta > 0$ *and* $d\lambda < 0$ *such that* $\int_{\theta_0}^{\overline{\theta}} \Delta f(\theta) d\theta$ *remains constant for some* $\theta_0 \in [\underline{\theta}, \overline{\theta})$*. There exists some* $\theta_1 \in (\theta_0, \overline{\theta})$ *such that* $\Delta(\theta)$ *decreases for* $\theta \in [\theta_0, \theta_1)$ *and increases for* $\theta \in (\theta_1, \overline{\theta}]$*.*

Part (i) of Lemma 3 shows that for the preferences to be convergent, the comparative advantage parameter, β, must be sufficiently small. Notably, if $\beta > 1 + 2\sqrt{1+\lambda}$, there will exist a range of θ such that the home country's non-cooperative tariff will increase more rapidly than the efficient tariff as the political shock increases. Part (ii) of Lemma 3 shows that an increase in β and a corresponding reduction in λ that keeps the average difference between importer and world preferences constant for $\theta > \theta_0$ will "rotate" the $\Delta(\theta)$ schedule around some point on the interval $\theta \in [\theta_0, \overline{\theta})$, making the preferences of the importer less convergent with that of the world as a whole on that interval.

The intuition for this result can be seen from the inverse of the foreign elasticity of export supply, which is a measure of the importer's market power.

$$\frac{1}{\varepsilon^*} = \frac{\lambda(1+\beta-2t)}{1+\beta}, \tag{20}$$

Market power is positive for $t < t^{pro}$, and is increasing in the home country's relative size, λ, and its degree of comparative disadvantage, β. Note however that the there is an important difference in the effect of these two parameters on market power. The marginal effect of country size on market power declines as t increases, because market power goes to 0 as the tariff approaches the prohibitive level and the prohibitive tariff is independent of market size. In contrast, the marginal effect of increases in β on market power increases as t rises because the prohibitive tariff is increasing in β. Since the difference between the non-cooperative tariff and the efficient tariff reflects the market power of the importing country, these observations indicate that preferences are more likely to be convergent when β is relatively low.

The necessary condition for an interior solution with a binding under a cap and escape agreement, (11), can be written in this case as

$$\frac{1-t^B}{1+t^B} = E\left[\theta|\theta^M(t^B) > \theta > \theta^B(t^B)\right] \tag{21}$$

The interpretation of this condition is similar to that in (18): the marginal welfare cost of increasing producer profits is equated to the expected political benefit over the set of political shocks for which the tariff is at the binding.

In order for a the solution to (21) to be a maximum, it must satisfy $\Lambda^M < 0$ from (12). For this case, the required condition is

$$\frac{2}{(1+t^B)^2} > \frac{\partial E\left[\theta|\theta^M > \theta > \theta^B)\right]}{\partial \theta^B}\frac{\partial \theta^B}{\partial t^B} + \frac{\partial E\left[\theta|\theta^M > \theta > \theta^B)\right]}{\partial \theta^M}\frac{\partial \theta^M}{\partial t^B}. \tag{22}$$

An increase in t^B raises both the upper and lower thresholds of the region where the tariff is at the binding. This contrasts while in the case without monitoring an increase in t^B only raises the lower threshold. Both of these effects increase the marginal benefit of raising the binding, making the conditions for an interior maximum more stringent than in the case without monitoring.

In order to address the existence of interior optima with both binding and overhang, we focus on the case where θ has a uniform distribution, which yields $E(\theta|\theta^M > \theta > \theta^B) = \frac{\theta^B+\theta^M}{2}$. The following Lemma establishes that if Nash and efficient tariffs are convergent at θ^M, or equivalently if $\beta < \tilde{\beta}(\theta^M)$, the second order condition (**??**) is violated.

Lemma 4 *Suppose that θ has a uniform distribution with $\underline{\theta} < \overline{\theta} < \theta^{\max}$ and the necessary condition is satisfied with $\underline{\theta} < \theta^B(t^B) < \theta^M(t^B) < \overline{\theta}$. This solution will fail to satisfy the second-order condition if $\beta < \tilde{\beta}(\theta^M)$.*

Lower values of β have two effects that make it more difficult to satisfy (12). One is that it makes the Nash tariff schedule less responsive to political shocks, which raises $\frac{\partial \theta^B}{\partial t^B}$. A second is that it raises the tariff binding associated with an interior solution for given θ^M, which reduces the left hand side of the inequality in (12). The proof

of Lemma 4 establishes that these two effects are sufficiently strong that the a local maximum cannot exist at θ^M if $\beta < \tilde{\beta}(\theta^M)$.

The following is an immediate implication of Lemma 4:

Proposition 7 *If preferences are globally convergent (i.e., if $\beta < \tilde{\beta}(0) \equiv 1+2\sqrt{1+\lambda}$) and θ has a uniform distribution, then escape and overhang do not coexist under an optimal cap-and-escape agreement.*

In other words, for a given country size, if the comparative advantage parameter is not too large, flexibility is provided either through overhang or an escape clause, but not both. The two forms of flexibility are sufficiently close substitutes that only one will be used in an agreement. Note that this result is also consistent with the corollary to Proposition 1, since it suggests that reducing convergence of preferences makes overhang less valuable as a flexibility mechanism.

3.2 A Monopolistic Competition Model

We can also derive payoff functions that satisfy Assumptions (1) and (2) from a two country, many sector model where sectoral competition is characterized by monopolistic competition. We assume I monopolitically competitive sectors, where sector $i = 1, ..., I$ consists of an exogenously given number n_i domestic firms and n_i^* foreign firms. Output of a firm in sector i requires c_i units of labor at home and c_i^* units of labor in the foreign country. We also assume a competitive numeraire sector in each country that produces using labor under c

Consumer preferences over the differentiated products are assumed to take the form

$$U = q_0 + \sum_i \left(a \int_0^{N_i} q_i(j)dj - \frac{g}{2} \int_0^{N_i} q_i(j)^2 dj - \frac{h}{2} \left(\int_0^{N_i} q_i(j)dj \right)^2 \right)$$

where $q_i(j)$ is the consumption of variety j from sector i, q_0 is consumption of the numeraire good, and $N_i = n_i + n_i^*$. We focus on a single differentiated products sector for simplicity in which the home country is an importer, and drop the sectoral subscript in the following discussion.

The demand for the product of a representative firm will be given by

$$q(j) = \frac{1}{g} \left(A(\bar{p}) - p(j) \right) \tag{23}$$

where $A(\bar{p}) = \frac{ag+hN\bar{p}}{g+hn}$ and $\bar{\bar{p}} = \frac{\int p(j)dj}{N}$ is the average price for the sector. Units of labor are chosen such that it requires one unit of labor per unit of the numeraire good in each country, and so the wage will equal 1 in each country. Assuming that markets are segmented and foreign producers face a specific tariff of t per unit, the

profit maximizing prices and profit levels of a representative firm of each type selling in the home country market will be

$$\begin{array}{ll} \tilde{p}(t) = \frac{A(\bar{p}(t))+c}{2} & \tilde{\pi}(t) = \frac{(A(\bar{p}(t))-c)^2}{4g} \\ \tilde{p}^*(t) = \frac{A(\bar{p}(t))+c^*+t}{2} & \tilde{\pi}^*(t) = \frac{(A(\bar{p}(t))-c^*-t)^2}{4g} \end{array} \tag{24}$$

Substituting for prices into into the demand function (23) yields the equilibrium sales of the respective firm types

$$\tilde{q}(t) = \frac{A(\bar{p}(t))-c}{2\gamma} \quad \tilde{q}^*(t) = \frac{A(\bar{p}(t))-c^*-t}{2\gamma} \tag{25}$$

An increase in the tariff shifts demand from foreign firms to domestic firms, and raises the price of both imported and domestically produced goods. Imports will be positive for tariffs below the prohibitive level, $t^{pro} = \frac{(\alpha-c^*)g+hn(c-c^*)}{2g+hn}$.

We assume that the political shock is captured as a weight on firm profits, as in the case of the perfectly competitive model. Home country welfare from home market activity will be

$$V(t,\theta) = S(\tilde{p}(t), \tilde{p}^*(t)) + (1+\theta)n\tilde{\pi}(t) + t\tilde{q}^*(t)$$

where $S(p, p^*)$ is consumer surplus. Totally differentiating this expression and using the properties of the pricing and profit functions Using $S_p(p, p^*) = -nd$, $S_{p^*}(p, p^*) = -n^*d^*$, (24) and (25) we can express the effect of an increase in tariff on welfare of

$$V_t(t,\theta) = n(1+2\theta)\left(\frac{\tilde{p}-c}{2}\right)\frac{d\tilde{q}}{dt} + n^*\left(t - \frac{\tilde{p}^*-c^*-t}{2}\right)\frac{d\tilde{q}^*}{dt} \quad \text{for } t < t^{pro} \tag{26}$$

The first term is the effect on home welfare of the increase in output of home firms as a result of an increase in the tariff. This term will be positive, because price exceeds marginal cost for home firms. The second term is the effect on home welfare of the decrease in the output of foreign firms as a result of the tariff. The second term shows that the reduction in the output of foreign firms due to the home tariff will raise home welfare as long as $p^* - c > 3t$. Home country welfare will be a quadratic in t, and will be strictly concave in t as long as θ is not too large. The above discussion established that $V_t(0,\theta) > 0$, so the non-cooperative home tariff will be positive.

World welfare will be $W(t,\theta) = V(t,\theta) + n^*\tilde{\pi}^*(t)$. Differentiating this expression yields *

$$W_t(t,\theta) = n(1+2\theta)\left(\frac{\tilde{p}-c}{2}\right)\frac{d\tilde{q}}{dt} + n^*\left(t + \frac{\tilde{p}^*-c^*-t}{2}\right)\frac{d\tilde{q}^*}{dt} \tag{27}$$

World welfare is also a quadratic in t, and will also be concave in t if θ is not too large. The magnitude of the spillover from the home tariff on foreign weflare is given by

$$V_t(t,\theta) - W_t(t,\theta) = n^*\left(\tilde{p}^*-c^*-t\right)\frac{d\tilde{q}^*}{dt} > 0 \text{ for } t < t^{pro}$$

The spillover is greater the larger the number of foreign firms in the home market and the An increase in the home tariff reduces foreign welfare, with the effect being

proportional to the markup of the foreign firm above the tariff inclusive marginal cost. The world welfare-maximizing tariff will be below the non-cooperative tariff of the importer due to the negative spillover from the home tariff.

4 Conclusions

Our analysis has shown how tariff bindings and contingent protection provide alternative means of introducing flexibility into trade agreements. Tariff overhang allows countries to make unilateral policy changes in response to political shocks, but has the disadvantage that the importing country will always choose a tariff that is higher than the first best tariff. As a result, tariff overhang will be used most extensively for countries that have relatively little market power. In contrast, contingent protection allows the imposition of tariffs that are efficient from a world point of view. However, it has the disadvantage of requiring the use of resources to verify the state.

Our results indicates that allowing contingent protection will result in a substitution of monitoring for tariff overhang if monitoring costs are sufficiently low. In particular, we showed that agreements will never involve both the use of contingent protection and tariff overhang if the preferences of the importer and the world are everywhere locally convergent. However, monitoring and overhang may coexist if preferences of the importer and the world are locally divergent. Our simulations also indicated that contingent has the lowest value for relatively small countries, since the use of tariff overhang will be a relatively more efficient mechanism for providing flexibility when terms of trade externalities are small.

Our analysis has highlighted the role of both market power and the convergence of preferences as determining the relative importance of tariff overhang and escape clauses in providing flexibility. Our analysis is the first to emphasize the role of preference convergence, which refers to whether the protectionist bias of importing countries increases or decreases as the magnitude of political shocks increase. Models which rely on the terms of trade externalities as the source of externalities in trade agreements will necessarily result in the convergence of preferences if political shocks are sufficiently large that they result in the imposition of prohibitive tariffs. However, our analysis shows that divergences of preferences can occur for smaller magnitudes of the political shocks and may play an important role in determining the value of tariff overhang. The role of preference divergence in other types of political economy models remains an area for future work.

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Appendix

Proof of Proposition 1: The Nash tariff satisfies $V_t(t^N, \theta) = -\mu$, so we have $\frac{\partial t^N}{\partial \mu} = -\frac{1}{V_{tt}(t^N,\theta)} > 0$ and $\frac{\partial \theta^B}{\partial \mu} = -\frac{1}{V_{t\theta}(t^N,\theta)} < 0$. Differentiating the necessary condition (5) yields

$$\frac{dt^B}{d\mu} = \frac{W_t(t^B, \theta^B) f(\theta^B) \frac{\partial \theta^B}{\partial \mu}}{\Lambda^B}$$

where Λ^B is given by (6). The result that $\frac{dt^B}{d\mu} < 0$ at an interior optimum follows from the fact that Λ^B, $\frac{\partial \theta^B}{\partial \mu}$ and $W_t(t^B, \theta^B)$ are all negative. At a corner solution, $\theta^B = \underline{\theta}$ and there is no effect on the optimal binding. The effect on the size of the boundary region is obtained from $\frac{d\theta^B}{d\mu} = \frac{\partial \theta^B}{\partial t^B} \frac{dt^B}{d\mu} + \frac{\partial \theta^B}{\partial \mu} < 0$.

The corollary follows from the observation that the comparative static result depends only on the behavior of $\theta^B(t)$ in the neighborhood of the binding.

Proof of Proposition 3: Totally differentiating the necessary condition (11) yields

$$\Lambda^M dt^B = W_t(t^B, \theta^B) f(\theta^B) \frac{\partial \theta^B}{\partial \mu} d\mu$$

where Λ^M is given by (12). The argument then proceeds as in Proposition 1 for the effects on θ^B and t^B. The decrease in θ^M follows from

Proof of Proposition **??**: The quadratic preferences yield tariff schedules $t^E(\theta) = \frac{a^W + \theta}{b^W}$ and $t^N(\theta) = \frac{a+\theta}{b}$. Solving for the thresholds of the binding region gives $\theta^M(t^B, c) = -a^W + b^W t^B + \sqrt{2b^W c}$ and $\theta^B(t^B) = -a + bt^B$. Substituting these results into (12), we have

$$\begin{aligned}\Lambda^M = -b^W \left(F(\theta^M(t^B)) - F(\theta^B(t^B))\right) + \sqrt{2b^W c} f(\theta^M(t^B)) b^W \\ - \left(a^W - a + (b - b^W)t^B\right) f(\theta^B(t^B)) b < 0\end{aligned}$$

The necessary condition for the optimal binding Under the assumption of a uniform distribution, this simplifies to

$$\underline{\left[\theta^M(t^B) - \theta^B(t^B) + \sqrt{2b^W c}\right]}$$

the necessary condition becomes

$$\left(a^W - b^W t^B\right)\left(\theta^M(t^B) - \theta^B(t^B)\right) + \frac{\theta^M(t^B)^2 - \theta^B(t^B)^2}{2} = 0,$$

which can be solved to has a solution an interior solution at

$$\Lambda^M = (b - b^W)(-a + a^W + (b - b^W)t^B) \tag{28}$$

Proof of Lemma 3 (i) Differentiation of (15) and (17) yields

$$\Delta'(\theta) = \frac{2\lambda\left[\beta^2(2-\theta)^2 - 2\beta\left(4-\theta^2\right) - 4(3+4\lambda-\theta) + \theta^2\right]}{(2-\theta)^2\left((\beta+1)(2-\theta)+4\lambda\right)^2}$$

The sign of this expression is determined by the sign of the bracketed expression in the numerator, which will be increasing in β because $2(2-\theta)\left[2(\beta-1)-(\beta+1)\theta\right] > 0$ for $\theta < \theta^{\max}$. The bracketed expression will equal 0 at $\beta = \frac{2+\theta+4\sqrt{1+\lambda}}{2-\theta}$.

ii) Let $d\lambda = -kd\beta < 0$, where $k = \left(\int_{\theta_0}^{\bar{\theta}} t_\beta^N(\theta)f(\theta)d\theta\right) / \left(\int_{\theta_0}^{\bar{\theta}} t_\lambda^N(\theta)f(\theta)d\theta\right) > 0$. Since $t_\beta^N(\theta)/t_\lambda^N(\theta)$ is continuous and increasing in θ by Lemma 1,$t_\beta^N(\theta_0)/t_\lambda^N(\theta_0) < k < t_\beta^N(\bar{\theta})/t_\lambda^N(\bar{\theta})$ and there will exist some $\theta_1 \in (\theta_0, \bar{\theta})$ such that $k = t_\beta^N(\theta)/t_\lambda^N(\theta)$. We then have $dt^N(\theta) = \left(t_\beta^N(\theta) - kt_\lambda^N(\theta)\right)d\beta > (<)\ 0$ for $\theta < (>)\theta_1$.

Proof. Define $J(t) = E(\theta|\theta \geq \theta^B(t)) - \frac{2t}{1+t}$, where $\theta^B(t) = \max\{\underline{\theta}.\frac{2t(1+\beta)+4\lambda(t-t^{pro})}{(1+t)(1+\beta)}\}$, which can be written as

$$J(t) = \begin{cases} \frac{\bar{\theta}+\gamma\underline{\theta}}{1+\gamma} - \frac{2t}{1+t}, & \text{if } t \in [0, t^N(\underline{\theta}]). \\ \frac{\bar{\theta}-\gamma\lambda\theta^{\max}-t\left(2-\bar{\theta}-\frac{4\gamma\lambda}{(1+\beta)}\right)}{(1+t)(1+\gamma)}, & \text{if } t \in [t^N(\underline{\theta}), t^N(\bar{\theta})]. \end{cases} \tag{A.1}$$

The necessary condition for optimal choice of binding, (**??**), will be satisfied if either $J(t) = 0$ or $t^B = t^N(\bar{\theta})$. If the solution to (**??**) also satisfies $(1-F(\theta^B(t))J'(t) - f(\theta^B(t))J(t)\frac{\partial\theta^B}{\partial t} < 0$, then it will be a local maximum. Since $J(t^N(\bar{\theta})) = \frac{\lambda(\bar{\theta}-\theta^{\max})}{(1+\lambda)} < 0$ for $\bar{\theta} < \theta^{\max}$,the solution with $t^B = t^N(\bar{\theta})$ will be a local minimum for $\bar{\theta} < \theta^{\max}$. A solution with $J(t) = 0$ will be a local maximum if $J'(t) < 0$. These observations, combined with the following Lemma (5), establish Proposition 1. ■

Lemma 5 *$J(t)$ is a continuous on $[0, t^N(\bar{\theta})]$, with $J(0) > 0$ and $J(t^N(\bar{\theta})) < 0$ for $\bar{\theta} < \theta^{\max}$.*

a) If $\gamma\lambda < 1$, then $J'(t) < 0$ and there exists a unique $t^B \in (0, t^N(\bar{\theta}))$ that maximizes expected welfare. If $\lambda \geq \tilde{\lambda} \equiv \frac{\bar{\theta}-\underline{\theta}}{(1+\gamma)\theta^{\max}-\bar{\theta}-\gamma\underline{\theta}}$, then $t^B = \frac{E(\theta)}{2-E(\theta)} \leq t^N(\underline{\theta}, \lambda)$. If $\lambda < \tilde{\lambda}$, then $t^B = \frac{\bar{\theta}-\gamma\lambda\theta^{\max}}{2-\bar{\theta}-4\gamma\lambda/(1+\beta)}(t^N(\underline{\theta}, \lambda), t^N(\bar{\theta}, \lambda)$

b) If $\gamma\lambda \geq 1$, then there exists a unique $t^B = \frac{\bar{\theta}-\gamma\lambda\theta^{\max}}{2-\bar{\theta}-4\gamma\lambda/(1+\beta)} \in (0, t^N(\underline{\theta}))$ that maximizes expected welfare.

Proof. From (A.1), $J(t)$ is continuous in t with $J'(t) = -\frac{2}{(1+t)^2}$ for $t < t^N(\underline{\theta})$ and $J'(t) = \frac{2(\gamma\lambda-1)}{(1+t)^2(1+\gamma)}$for $t > t^N(\underline{\theta})$.

(a) Since $J(0) = E(\theta) > 0$ and $J(t^N(\bar{\theta})) < 0$ as established above, there will exist a unique t^B such that $J(t^B) = 0$ if $\gamma\lambda < 1$. This solution will be a global maximum since $J'(t^B) < 0$. A corner binding with no overhang will exist if $t^B = \frac{E(\theta)}{2-E(\theta)} \leq t^N(\underline{\theta}, \lambda)$,

which is shown to hold for $\lambda \geq \tilde{\lambda}$ using (15). For $\lambda < \tilde{\lambda}$, the optimal binding is obtained by solving $J'(t^B) = 0$.

(b) If $\gamma\lambda > 1$, $J'(t) > 0$ for $t \in (t^N(\underline{\theta}), t^N(\overline{\theta}))$ and there cannot exist a local maximum on this interval. This yields two possibilities: a corner solution with no overhang or a corner solution with an unbound tariff (i.e. $t^B \geq t^N(\overline{\theta})$). Since $\tilde{\lambda} < \frac{1}{\gamma}$ when $\overline{\theta} < \theta^{\max}$, $\gamma\lambda > 1$ implies $\lambda > \tilde{\lambda}$ and there will exist a local maximum with no overhang, $t^B \in (0, t^N(\underline{\theta}))$, by the arguments in (a). To show that there cannot be an optimum at a corner solution with an unbound tariff, we show that $EW[t^B]$ is decreasing on $(t^N(\underline{\theta}), t^N(\overline{\theta}))$. Since sign $EW'[t^B]$ =sign $J(t)$, it is sufficient to show that $J(t) < 0$ on that interval. We have established that $J(t)$ is continuous with $J(t^N(\underline{\theta})) < 0$, $J(t^N(\overline{\underline{\theta}})) < 0$, and $J'(t) > 0$ for $t \in (t^N(\underline{\theta}), t^N(\overline{\theta}))$. Therefore we must have $J(t) < 0$ on the interval. ■